Water-Cooled Process Chiller (CHILLER) Equipment Manual and SOP

VINDHYA PRECISION WORKS ## STANDARD OPERATING PROCEDURE AND EQUIPMENT MANUAL Asset Class: CHILLER (Water-Cooled Process Chiller) Document ID: SOP-MNT-CHILLER-042 Revision: 4.1 Effective Date: 14 January 2025 Owner: Maintenance Engineering Lead

1. SCOPE AND RESPONSIBILITY

1.1 This document governs all preventive maintenance, corrective maintenance, condition-monitoring thresholds, and fault-resolution procedures for the Water-Cooled Process Chiller asset class (CHILLER) installed at Vindhya Precision Works.

1.2 Primary maintenance responsibility is assigned to the Mechanical trade, with mandatory Electrical trade support for all control, power, and interlock verifications.

2. NORMAL OPERATING RANGES AND CONDITION MONITORING THRESHOLDS

2.1 Technicians shall inspect the chiller against the operational limits detailed in Table 2.1. Any reading outside the normal healthy range but below the warning threshold requires a logged observation within the Computerized Maintenance Management System (CMMS).

Table 2.1: CHILLER Operating Parameters and Alarm Setpoints

Parameter	Healthy Operating Range	Warning Setpoint	Trip / Interlock Setpoint	Action on Trip
Vibration (RMS)	1.2 to 4.0 mm/s	5.3 mm/s RMS	8.2 mm/s RMS	Immediate compressor coast-down; Lockout Code F-02
Surface Temperature	38 °C to 66 °C	80 °C	95 °C	Controlled shutdown via refrigerant bypass; Lockout Code F-03
Operating Pressure	2.8 to 3.2 bar (3.0 bar nominal)	2.4 bar (Low Pressure)	1.9 bar	Immediate compressor trip; Lockout Code F-01
Motor Current	78 A to 92 A (85 A nominal)	98 A (105% FL)	110 A (125% FL)	Overcurrent breaker trip; Lockout Code F-04
Chilled Water Outlet	6.5 °C to 7.5 °C (7.0 °C nominal)	5.0 °C / 9.0 °C	3.5 °C / 10.5 °C	Process interlock alarm; Chiller modulation halt

3. PREVENTIVE MAINTENANCE SCHEDULE

3.1 The base preventive maintenance interval for asset class CHILLER is 3000 running hours or every 12 calendar months, whichever occurs first.

3.2 Running-Hour Tasks:
* 3000 Hours (PM-1): Inspect and clean condenser tubes; test oil separator differential pressure; replace fluid filters; verify refrigerant charge via subcooling calculation; inspect electrical terminal torque; test safety pressure switches.
* 6000 Hours (PM-2): Perform PM-1 tasks plus: service compressor oil pump; calibrate all temperature and pressure transducers; inspect compressor slide valve mechanism; test vibration sensor loop integrity.
* 12000 Hours (PM-3): Perform PM-2 tasks plus: overhaul compressor internal bearings; inspect evaporator tube bundle; replace all dynamic shaft seals and gaskets; perform non-destructive examination (NDE) of refrigerant containment pressure vessels.

4. FAULT-CODE TABLE AND CORRECTIVE ACTIONS

4.1 When the control panel registers a fault code, technicians shall execute the diagnostic and corrective actions specified in Table 4.1.

Table 4.1: CHILLER Fault Codes and Remediation

Fault Code	Named Fault Condition	Probable Cause	Corrective Action
F-01	Refrigerant Low Pressure Trip	Restricted liquid line, condenser fouling, or minor refrigerant charge loss.	Inspect sight glass for bubbles; check for oil traces at joints; leak-test using electronic detector; recover, evacuate, and recharge if charge is deficient by >10%. Replace part VPW-P-00022 (Per filter) if differential pressure exceeds 0.7 bar.
F-02	Compressor High Vibration	Coupling misalignment, worn internal bearings, or dynamic rotor unbalance.	Measure bearing housing vibration. If RMS velocity exceeds 8.2 mm/s, isolate unit, check alignment, and replace part VPW-P-00015 (Down coupling) and part VPW-P-00010 (We bearing).
F-03	Compressor High Surface Temperature	Inadequate cooling water flow, scaling on condenser tubes, or oil cooler starvation.	Verify condenser cooling water flow rate ($\ge 120 \text{ m}^3\text{/h}$); backflush condenser tubes; inspect oil injection solenoid valve part VPW-P-00043 (Cause valve) for correct actuation.
F-04	Motor Overcurrent Trip	Phase imbalance, high ambient plant temperature, or internal mechanical binding.	Measure line-to-line resistance and voltage balance; check starter contactor condition; verify motor winding thermistor resistance ($\le 2.7\text{ k}\Omega$ at normal operating temp).
F-05	Slide Valve Positioning Error	Hydraulic actuator failure, internal linkage binding, or feedback sensor drift.	Inspect hydraulic control manifold; verify actuator pressure (nominal 18 bar); replace part VPW-P-00012 (Live valve) if solenoid spool is stuck; recalibrate feedback transducer part VPW-P-00027 (Program sensor).
F-06	Sensor Loop Open / Out of Range	Damaged signal cable, water ingress into terminal box, or transducer failure.	Verify terminal strip continuity; check 24VDC loop power supply; if sensor reads outside physical limits, replace part VPW-P-00014 (End sensor) and inspect connector sealing gland.

5. CONSUMABLE AND SPARE PARTS LIST

5.1 Technicians shall only use the exact inventory part numbers listed in Table 5.1 for maintenance and repair of asset class CHILLER. No substitutions are permitted.

Table 5.1: Approved CHILLER Spare Parts Inventory

Part Number	Description	Standard Replacement Interval
VPW-P-00010	We bearing	12,000 running hours (PM-3)
VPW-P-00012	Live valve	As required by fault F-05
VPW-P-00014	End sensor	As required by fault F-06
VPW-P-00015	Down coupling	12,000 running hours or upon high vibration
VPW-P-00022	Per filter	3,000 running hours (Every PM-1)
VPW-P-00027	Program sensor	6,000 running hours (Every PM-2)
VPW-P-00041	Difficult seal	12,000 running hours (Every PM-3)
VPW-P-00043	Cause valve	As required by fault F-03

6. TORQUE SPECIFICATIONS

6.1 Technicians shall apply the torque values specified in Table 6.1 during reassembly. Calibrated torque wrenches are mandatory for all bolting operations.

Table 6.1: Fastener Torque Limits

Fastener Location	Thread Size	Target Torque	Lubrication Status
Compressor Terminal Lug Nuts	M10	28 Nm	Clean, dry threads
Refrigerant Flange Bolting (High Pressure)	M12	65 Nm	Anti-seize compound applied
Oil Line Compression Fittings	3/4-16 UNF	45 Nm	Light mineral oil on threads
Coupling Clamping Bolts	M8	32 Nm	Clean, dry threads
Bearing Housing Retaining Bolts	M10	40 Nm	Thread-locking fluid (Medium strength)

7. SAFETY-CRITICAL CORRECTIVE ACTIONS AND LOCKOUT / PERMIT REQUIREMENTS

7.1 Hazardous Energy Sources: Prior to initiating any corrective maintenance on asset class CHILLER, technicians must isolate the following two primary hazardous energy sources:
1. Electrical Energy: 415V AC 3-Phase supply via main feeder breaker in Motor Control Center (MCC) Panel CH-04.
2. Refrigerant Pressure: High-pressure liquid and vapor R-134a/R-513A containment within the compressor, condenser, and evaporator vessels.

7.2 Safety-Critical Corrective Actions: The following corrective actions are designated as safety-critical and strictly require an approved Plant Lockout/Tagout (LOTO) and a formal Confined Space / Hot Work / Pressure Vessel Opening Permit signed by the shift safety officer before work begins:
* Action A: Replacement or internal servicing of part VPW-P-00010 (We bearing) or compressor disassembly.
* Action B: Opening of any refrigerant-containing circuit, line, or vessel requiring evacuation, refrigerant recovery, and replacement of part VPW-P-00022 (Per filter) or part VPW-P-00041 (Difficult seal).
* Action C: Maintenance, removal, or replacement of high-pressure hydraulic or solenoid control valves, specifically part VPW-P-00012 (Live valve) or part VPW-P-00043 (Cause valve).
* Action D: Electrical busbar inspection, drive-inverter replacement, or main power terminal re-termination inside the compressor terminal enclosure.

7.3 No technician shall bypass safety interlocks, pressure relief valves, or vibration trip loops without written authorization from the Maintenance Engineering Lead and the execution of a temporary operating procedure document.